Year 10 Higher

REVISION PACK

Number 1

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Number 1 · Working with fractions (Lesson 1.1)

Q	Working	Answer
adding fractions		
1.1	$\frac{12}{15} + \frac{10}{15} = \frac{22}{15}$ $= 1\frac{7}{15}$	$1\frac{7}{15}$
1.2	$\frac{20}{24} + \frac{9}{24} = \frac{29}{24}$ $= 1\frac{5}{24}$	$1\frac{5}{24}$
1.3	$\frac{9}{12} + \frac{8}{12} = \frac{17}{12}$ $= 1\frac{5}{12}$	$1\frac{5}{12}$
1.4	$\frac{10}{14} + \frac{7}{14} = \frac{17}{14}$ $= 1\frac{3}{14}$	$1\frac{3}{14}$
1.5	$\frac{21}{24} + \frac{8}{24} = \frac{29}{24}$ $= 1\frac{5}{24}$	$1\frac{5}{24}$
1.6	$\frac{10}{18} + \frac{15}{18} = \frac{25}{18}$ $= 1\frac{7}{18}$	$1\frac{7}{18}$
multiplying mixed numbers		
1.7	$\frac{3}{2} \times \frac{8}{3}$ $= \frac{24}{6}$ $= 4$	4

1.8	$\frac{10}{3} \times \frac{8}{5}$ $= \frac{80}{15}$ $= \frac{16}{3}$ $= 5\frac{1}{3}$	$5\frac{1}{3}$
1.9	$\frac{9}{4} \times \frac{11}{6}$ $= \frac{99}{24}$ $= \frac{33}{8}$ $= 4\frac{1}{8}$	$4\frac{1}{8}$
1.10	$\frac{7}{4} \times \frac{12}{5}$ $= \frac{84}{20}$ $= \frac{21}{5}$ $= 4\frac{1}{5}$	$4\frac{1}{5}$
1.11	$\frac{8}{3} \times \frac{9}{8}$ $= \frac{72}{24}$ $= 3$	3
1.12	$\frac{15}{4} \times \frac{5}{3}$ $= \frac{75}{12}$ $= \frac{25}{4}$ $= 6\frac{1}{4}$	$6\frac{1}{4}$
dividing mixed numbers		

1.13	$\frac{9}{2} \div \frac{6}{5}$ $= \frac{9}{2} \times \frac{5}{6}$ $= \frac{45}{12}$ $= \frac{15}{4}$ $= 3\frac{3}{4}$	$3\frac{3}{4}$
1.14	$\frac{8}{3} \div \frac{4}{3}$ $= \frac{8}{3} \times \frac{3}{4}$ $= \frac{24}{12}$ $= 2$	2
1.15	$\frac{15}{4} \div \frac{5}{2}$ $= \frac{15}{4} \times \frac{2}{5}$ $= \frac{30}{20}$ $= \frac{3}{2}$ $= 1\frac{1}{2}$	$1\frac{1}{2}$
1.16	$\frac{21}{4} \div \frac{7}{4}$ $= \frac{21}{4} \times \frac{4}{7}$ $= \frac{84}{28}$ $= 3$	3
1.17	$\frac{12}{5} \div \frac{6}{5}$ $= \frac{12}{5} \times \frac{5}{6}$ $= \frac{60}{30}$ $= 2$	2

1.18	$\frac{25}{6} \div \frac{5}{3}$ $= \frac{25}{6} \times \frac{3}{5}$ $= \frac{75}{30}$ $= \frac{5}{2}$ $= 2\frac{1}{2}$	$2\frac{1}{2}$
subtracting with a mixed number		
1.19	$\frac{7}{3} - \frac{5}{6}$ $\frac{14}{6} - \frac{5}{6} = \frac{9}{6}$ $= \frac{3}{2}$ $= 1\frac{1}{2}$	$1\frac{1}{2}$
1.20	$\frac{8}{5} - \frac{3}{4}$ $\frac{32}{20} - \frac{15}{20} = \frac{17}{20}$	$\frac{17}{20}$
1.21	$\frac{13}{4} - \frac{2}{3}$ $\frac{39}{12} - \frac{8}{12} = \frac{31}{12}$ $= 2\frac{7}{12}$	$2\frac{7}{12}$
1.22	$\frac{12}{5} - \frac{5}{6}$ $\frac{72}{30} - \frac{25}{30} = \frac{47}{30}$ $= 1\frac{17}{30}$	$1\frac{17}{30}$
1.23	$\frac{11}{6} - \frac{3}{4}$ $\frac{22}{12} - \frac{9}{12} = \frac{13}{12}$ $= 1\frac{1}{12}$	$1\frac{1}{12}$

1.24	$\frac{5}{2} - \frac{4}{5}$ $\frac{25}{10} - \frac{8}{10} = \frac{17}{10}$ $= 1\frac{7}{10}$	$1\frac{7}{10}$
fraction problems		
1.25	$1 - \frac{2}{5} = \frac{3}{5}$ $\frac{1}{3} \times \frac{3}{5} = \frac{1}{5}$ $\frac{3}{5} - \frac{1}{5} = \frac{2}{5}$	$\frac{2}{5}$
1.26	$1 - \frac{1}{4} = \frac{3}{4}$ $\frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$ $\frac{3}{4} - \frac{1}{2} = \frac{1}{4}$	$\frac{1}{4}$
1.27	$1 - \frac{3}{8} = \frac{5}{8}$ $\frac{1}{5} \times \frac{5}{8} = \frac{1}{8}$ $\frac{5}{8} - \frac{1}{8} = \frac{1}{2}$	$\frac{1}{2}$
1.28	$1 - \frac{1}{6} = \frac{5}{6}$ $\frac{3}{5} \times \frac{5}{6} = \frac{1}{2}$ $\frac{5}{6} - \frac{1}{2} = \frac{1}{3}$	$\frac{1}{3}$
1.29	$1 - \frac{3}{7} = \frac{4}{7}$ $\frac{1}{2} \times \frac{4}{7} = \frac{2}{7}$ $\frac{4}{7} - \frac{2}{7} = \frac{2}{7}$	$\frac{2}{7}$
1.30	$1 - \frac{2}{9} = \frac{7}{9}$ $\frac{3}{7} \times \frac{7}{9} = \frac{1}{3}$ $\frac{7}{9} - \frac{1}{3} = \frac{4}{9}$	$\frac{4}{9}$

Number 1 · Order of operations and rounding (Lessons 1.2, 1.3)		
Q	**Working**	**Answer**
order of operations		
2.1	**(a)** $5^2 = 25$ $2 \times 25 = 50$ $7 + 50 = 57$ **(b)** $(3 + 5) \times 4 - 2 = 30$	**(a) 57** **(b)** $(3 + 5) \times 4 - 2 = 30$
2.2	**(a)** $2^3 = 8$ $3 \times 8 = 24$ $20 - 24 = -4$ **(b)** $2 \times (6 + 3) - 1 = 17$	**(a) −4** **(b)** $2 \times (6 + 3) - 1 = 17$
2.3	**(a)** $3^2 = 9$ $18 \div 9 = 2$ $6 + 2 = 8$ **(b)** $4 + 8 \times (2 + 1) = 28$	**(a) 8** **(b)** $4 + 8 \times (2 + 1) = 28$
2.4	**(a)** $2^3 = 8$ $4 \times 8 = 32$ $9 + 32 = 41$ **(b)** $5 \times (4 - 1) + 2 = 17$	**(a) 41** **(b)** $5 \times (4 - 1) + 2 = 17$
2.5	**(a)** $2^2 = 4$ $6 \times 4 = 24$ $3 + 24 = 27$ **(b)** $(10 - 2) \times 3 + 1 = 25$	**(a) 27** **(b)** $(10 - 2) \times 3 + 1 = 25$
2.6	**(a)** $3^2 = 9$ $5 \times 9 = 45$ $40 - 45 = -5$ **(b)** $(6 + 2) \times 7 - 3 = 53$	**(a) −5** **(b)** $(6 + 2) \times 7 - 3 = 53$
rounding		
2.7		**(a) 275.8** **(b) 280**

2.8		**(a) 1093.5** **(b) 1100**
2.9		**(a) 48.1** **(b) 48**
2.10		**(a) 736.3** **(b) 740**
2.11		**(a) 5281.6** **(b) 5300**
2.12		**(a) 20.0** **(b) 20**